

C#: Converting String to Sbyte*

Asked 14 years, 8 months ago Modified 14 years, 8 months ago Viewed 23k times



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My C# code uses a Managed C++ Wrapper. To make a new object of this Wrapper's type, I need to convert String's to Sbyte*'s. A few StackOverflow.com posts discussed how to convert String to byte[], as well as byte[] to sbyte[], but not String to sbyte*.

msdn.social.com offers advice on how to convert a byte array to a string:

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```
> // convert String to Sbyte*
> string str = "The quick brown, fox jumped over the gentleman.";
>
> System.Text.ASCIIEncoding encoding = new
> System.Text.ASCIIEncoding();
>
> Byte[] bytes = encoding.GetBytes(str);
```

However, "bytes" is not of type sbyte*. My following attempts to convert bytes to sbyte* failed:

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```
1. Convert.ToSbyte(bytes);
2. cast: (sbyte*) bytes;
```

Please advise me on how to convert a C# string to an sbyte*.

Also, please talk about any side effects from introducing sbyte*, which I believe is unsafe code.

Thanks, Kevin

c#

unmanaged

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asked Apr 14, 2011 at 16:00



Kevin Meredith

42.1k ● 67 ● 224 ● 390

3 Answers

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Hmmm how about something like this:

(didn't test it, don't give me -1 if it doesn't work, I just believe that it should :))



```
string str = "The quick brown fox jumped over the gentleman.";
byte[] bytes = Encoding.ASCII.GetBytes(str);

unsafe
{
    fixed (byte* p = bytes)
    {
        sbyte* sp = (sbyte*)p;
        //SP is now what you want
    }
}
```

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answered Apr 14, 2011 at 16:11



Cipi

11.4k ● 9 ● 50 ● 60

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Kevin Meredith Over a year ago

Thanks, Cipi. I'll go try this out.



0



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You can do that way:



```
sbyte[] sbytes = Array.ConvertAll(bytes, q => Convert.ToSByte(q));
```

2 Comments ▾

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answered Apr 14, 2011 at 16:07

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Felice Pollano

33.4k ● 10 ● 77 ● 126



Kevin Meredith

Over a year ago

Once I convert to sbytes[], how do I convert from sbytes[] to sbyte*? A simple cast does not work. Ty



Reply



Darkgaze

Over a year ago

@KevinMeredith the first item in the array is the sbyte* -> sbyte* b = &sbytes[0]



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sbyte[] and sbyte* are almost the same thing (almost)

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sbyte[] str1; sbyte* str2;



&str1[0] is a pointer to the first element of the array of chars str2 is a pointer to a char, which presumably has a bunch of consecutive chars following it, followed by a null terminator



if you use str1 as an array, you know the length without a null terminator. but if you want to pass str1 to a string api that requires sbyte*, you use &str1[0] to turn it into a sbyte*, and then you stripped away array length information, so you have to make sure you're null terminated.

Cipi's answer shows you how to convert an array to a pointer, C# makes it hard to use pointers, on purpose. in C or C++ arrays and pointers are similar.

<http://www.lysator.liu.se/c/c-faq/c-2.html>

<http://pw1.netcom.com/~tjensen/ptr/pointers.htm>

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edited Apr 14, 2011 at 17:59

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answered Apr 14, 2011 at 17:54



Dustin Getz

22k ● 15 ● 87 ● 132

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